

Conversation State and Memory

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of conversation state and retention.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 20: Conversation State and Memory

Products choose what context to keep, summarize, forget, or retrieve.



CREATOR MISSION

Design memory rules for a tutoring conversation.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Tools and API contracts. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

By the end, I can explain conversation state and memory using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use conversation state and memory to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Conversation state stores facts or prior interactions for later use. It needs scope, provenance, correction and retention rules rather than indefinite accumulation.

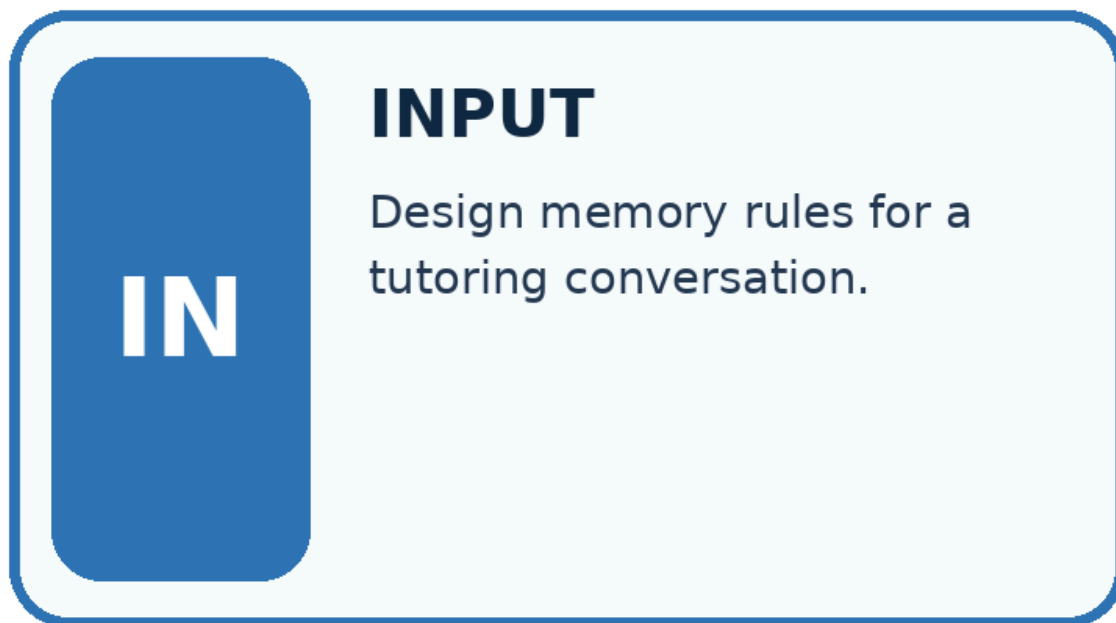
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Products choose what context to keep, summarize, forget, or retrieve.



OUT

EVIDENCE

Design memory rules for a tutoring conversation.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

This classroom assistant keeps an approved room ID only for the current session. A later correction replaces the active value but retains an audit event until session end. Do not persist private text merely because it appeared in conversation.

Conversation state and retention — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Turn 1 selects room A; turn 2 corrects to room B. A later answer uses A from an older summary.

Check or calculation: This classroom assistant keeps an approved room ID only for the current session. A later correction replaces the active value but retains an audit event until session end. Do not persist private text merely because it appeared in conversation.

Conclusion and limit: Active state should be B under the correction rule. The summary is stale; regenerate or invalidate it and preserve the correction audit within the declared session lifetime.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: The next session asks “summarise my book” without naming one.

Prediction: Ask which book the learner wants.

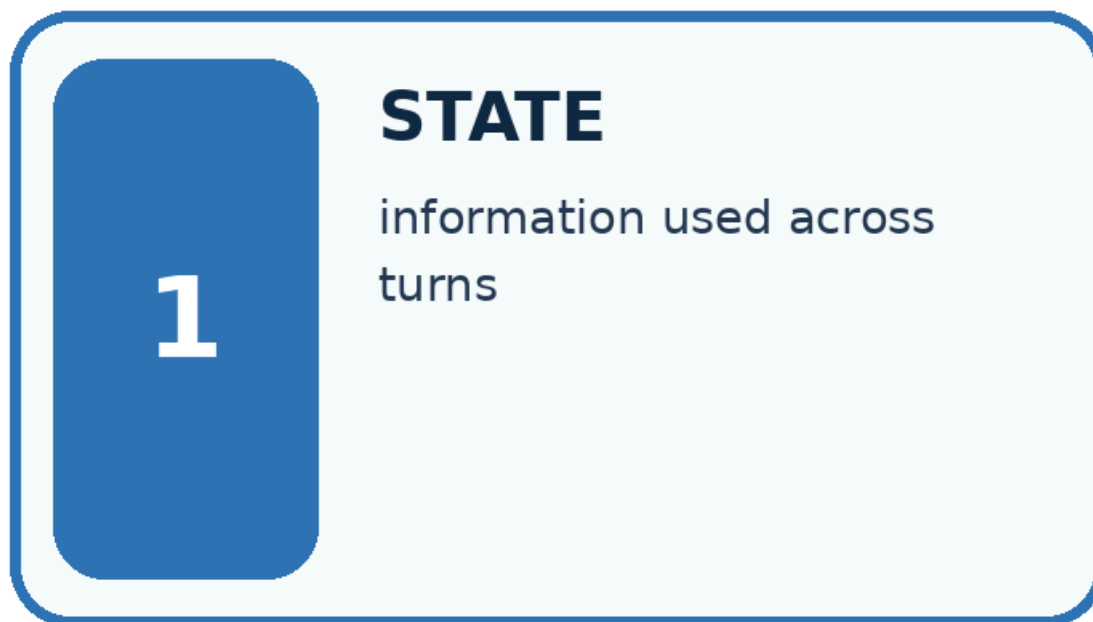
Reason: The declared policy clears session state, so yesterday’s choice is unavailable.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

SESSION MEMORY

temporary context during
one interaction

3

PERSISTENT MEMORY

stored information across
sessions

4

RETENTION RULE

when information is deleted

Word · My own example · Not this / boundary

Word: state

My own example: The current values and information that determine what a process does next.

Not this / boundary: _____

Word: session memory

My own example: Information maintained during the current interaction, without assuming it persists afterwards.

Not this / boundary: _____

Word: persistent memory

My own example: Information retained beyond the current session by an explicit storage mechanism.

Not this / boundary: _____

Word: retention rule

My own example: A policy specifying what information is kept, for how long and when it is deleted.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Session state

Meaning: Information retained within a defined interaction lifetime

Word: Correction event

Meaning: A record updating an earlier active value

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 20 • WORKED EVIDENCE

- 1 Keep current topic and learner's chosen difficulty for the session.
- 2 Summarize progress without storing full private conversation.
- 3 Forget sensitive or unnecessary details.
- 4 Retrieve approved study notes only when the learner requests that topic.

1. Keep current topic and learner's chosen difficulty for the session.
2. Summarize progress without storing full private conversation.
3. Forget sensitive or unnecessary details.
4. Retrieve approved study notes only when the learner requests that topic.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Turn 1 selects room A; turn 2 corrects to room B. A later answer uses A from an older summary.

Active state should be B under the correction rule. The summary is stale; regenerate or invalidate it and preserve the correction audit within the declared session lifetime.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Conversation State and Memory

Keep a reading conversation consistent when the learner changes the selected book.

The facts we will use

In this toy session, “choose Cedar” sets selected_book=Cedar; “change to Maple” replaces it. Session state is cleared on exit.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Conversation State and Memory

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First choice

Store Cedar as this session's selected book.



2. Correct choice

Replace Cedar with Maple; do not append an equally active second choice.



3. End session

Clear selected_book. A new session begins with no selected book.

CHANGE ONE THING

The next session asks "summarise my book" without naming one.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. First choice

What happens and why: Store Cedar as this session's selected book.

Step: 2. Correct choice

What happens and why: Replace Cedar with Maple; do not append an equally active second choice.

Step: 3. End session

What happens and why: Clear selected_book. A new session begins with no selected book.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Clear selected_book. A new session begins with no selected book. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Conversation state is an application record; it is not evidence that the model permanently learned the learner's preferences.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Remembering everything always creates a better assistant. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Remembered statements are not automatically current or verified; retention should follow the stated purpose and policy.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Conversation state and retention”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Conversation state and retention. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Sort information cards into keep, summarize, forget, or retrieve; justify purpose, permission, and retention time.

Sort information cards into keep, summarize, forget, or retrieve; justify purpose, permission, and retention time.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Store minimal session state and make persistent memory optional, visible, and erasable.

THIS WEEK'S CHECKPOINT Store minimal session state and make persistent memory optional, visible, and erasable.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Design memory rules for a tutoring conversation.

CHALLENGE Design memory rules for a tutoring conversation.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

State stores topic=plants and reading_level=beginner. User requests advanced explanation.

The question

What changes?

Worked explanation

Update reading level, retain topic unless changed. Keep only necessary state with correction/reset. Context does not guarantee permanent memory.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Update only the state the user changed

Evidence or stage · Value or result

Evidence or stage: Before

Value or result: topic=plants, level=beginner

Evidence or stage: Request

Value or result: Use an advanced explanation

Evidence or stage: After

Value or result: topic=plants, level=advanced

Evidence or stage: Reset

Value or result: Clear stored fields when requested

State stores topic=plants and reading_level=beginner. User requests advanced explanation.

What changes?

Update reading level, retain topic unless changed. Keep only necessary state with correction/reset.

Context does not guarantee permanent memory.

14. Try a new case independently

Independent case: Apply a new case independently

A session stores lab L1 then corrects to L2; session ends. A new session has no room input. State the active values at each stage.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A learner corrects their project topic from oceans to forests. The next reply still uses oceans. What state should change, and what should not be saved unnecessarily?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Store minimal session state and make persistent memory optional, visible, and erasable.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

The conversation state is topic=plants, level=Grade 8. The user says 'Keep the topic, explain for Grade 6'. Write the updated state and a reset behaviour.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.